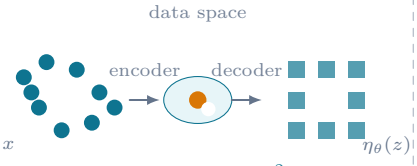
Three model families use three different geometries

VAE



$q_\phi(z \mid x) = \mathcal{N}(\mu, \sigma^2)$

$z = \mu + \sigma \odot \varepsilon$

train: $\log p_\theta(x \mid z) - \mathrm{KL}$

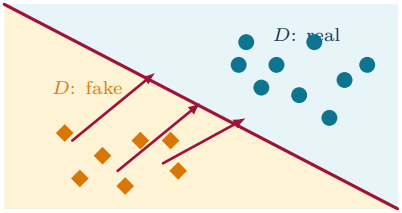
generate: $z \sim p(z),\ x \sim p_\theta(\cdot \mid z)$

ε enters differentiably

GAN

real x

$G(z)$

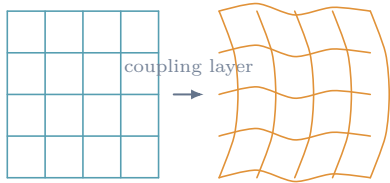


generator gradient moves fake samples toward regions judged real

alternate D update $\leftrightarrow$ G update

RealNVP

$z_a = x_a \qquad z_\beta = x_\beta\, e^{s(x_a)} + t(x_a)$



inverse is explicit

local area change $= \exp \sum_j s_j$

exact likelihood from the Jacobian